When Colleges Come Closer: The Effects of Local Higher Education Expansion on Educational Outcomes in India

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Abstract

India has experienced rapid growth in higher education over the past two decades. However, access remains uneven across regions. This paper examines the impact of Model Degree Colleges established under the Rashtriya Uchchatar Shiksha Abhiyan (RUSA), a centrally sponsored program launched in 2013 to expand access to higher education in underserved areas. The program targeted Educationally Backward Districts (EBDs) with low college availability, low female enrollment, and high shares of marginalized populations. Using Periodic Labour Force Survey (PLFS) data from 2017 to 2023, I exploit variation in the timing of college establishment across districts and estimate the effects using a staggered difference-in-differences design. I find that opening a Model Degree College increases educational attainment. Years of formal education increase by 0.63 years for individuals who were 18 when the college opened, with the gains driven by men, while women show positive but statistically insignificant effects. The effects are also larger among individuals from Scheduled Caste and Scheduled Tribe (SC/ST) groups, who gain 0.88 years, and Other Backward Class (OBC) groups, who gain 0.84 years. I also find suggestive evidence that college opening increases graduation and delays marriage among younger women.

Keywords: higher education, India, human capital

JEL Code: I23, I26, H52

1 Introduction

Access to higher education is unevenly distributed within countries as well as across them. Colleges tend to be concentrated in a few urban areas, so continuing education after secondary school often means leaving home, and the cost of relocating can discourage students

from enrolling. High-income countries have addressed this in part through community colleges, sited outside major urban centers, which serve students who need to remain close to home (Mountjoy, 2022).

The problem is more acute in developing countries. Higher education systems are smaller, and many institutions date to the colonial period and are located in cities, and historically served a narrow elite (World Bank, 2000). Constraints on mobility are also more binding, so distance weighs most heavily on disadvantaged students (Borker et al., 2021; Muralidharan and Prakash, 2017). Enrollment reflects these gaps: the gross enrollment ratio (GER) in higher education averaged 39 percent across low- and middle-income countries in 2024, compared with 82 percent in high-income countries (UNESCO, 2026).

One policy response is to expand the supply of public colleges in underserved regions. Whether this raises enrollment depends on how much distance matters relative to other barriers, such as financial pressures and social norms, which may limit college attendance even when a college is nearby. This paper asks to what extent opening public colleges improves educational outcomes, and for whom.

India provides an important setting to study this question. It has the third-largest higher education system in the world, yet access is highly unequal across regions. Participation is low: the GER was only 28.4 percent in 2021–22 (AISHE, 2022), below the average for low- and middle-income economies.[1] Much of India's expansion has come from private colleges concentrated in a few districts, leaving many areas with far fewer colleges than others. To address these gaps, the Government of India launched the Rashtriya Uchchatar Shiksha Abhiyan (RUSA) in 2013, a centrally sponsored program to improve access, quality, and equity in higher education. A key component was the establishment of Model Degree Colleges in Educationally Backward Districts (EBDs), selected on the basis of low college availability, low female enrollment, and a high share of marginalized populations.

I study the effect of these colleges on the college-age population in the districts that

[1]The UNESCO Institute for Statistics reports a GER in higher education of 31.6 percent for India in 2021, compared with 35.8 percent across low- and middle-income countries (UNESCO, 2026).

received them. Treatment is identified from project approvals by a designated Ministry of Education committee, and college establishment is verified using institution-level data from the All India Survey of Higher Education (AISHE). Educational attainment and labor market outcomes are measured using the Periodic Labour Force Survey (PLFS). I compare districts that received a college with eligible districts that did not, exploiting variation in both treatment status and timing. The analysis focuses on colleges established between 2018 and 2020, for which the 2017–2023 PLFS provides observations both before and after establishment. Because colleges opened in different years, I use a staggered difference-in-differences design, estimated with the two-stage estimator of Gardner (2022).

The effect of a new college depends on who is still making schooling decisions when it opens. I focus on individuals who were 18 at establishment, the typical age of college entry in India, and also examine those who were 17. Those who were 19 had likely already decided, so I use them as a placebo group.

The benefits may also be unevenly distributed within a district. Because the program targets districts rather than individuals, better-off students may gain the most. I therefore examine effects separately for Scheduled Castes and Scheduled Tribes (SC/ST), the historically marginalized groups, and for Other Backward Classes (OBC). Women face additional constraints, as for them the typical age of college entry coincides with the legal age of marriage, 18, and marriage is typically patrilocal, with women moving to their husband's household. Many women may therefore leave education before a college opens, limiting their ability to benefit from it.

I find that expanding access to local public colleges increases educational attainment. For individuals who were 18 when a college opened in their district, years of formal education increase by 0.63 years. The gains are also larger among historically disadvantaged groups, with educational attainment increasing by 0.88 years among Scheduled Caste and Scheduled Tribe (SC/ST) individuals and 0.84 years among Other Backward Class (OBC) individuals. The effects are driven primarily by men, whose educational attainment increases by 0.93

years. These results suggest that better-off students do not capture the gains, although women benefit less than men.

I then extend the analysis to individuals who were 17 years old when the college opened. The estimated effect is positive but smaller, with years of formal education increasing by 0.39 years. As a placebo test, I examine individuals who were 19 years old at establishment, for whom the estimated effect is small and not statistically significant. The pattern across cohorts is consistent with the main effect being concentrated among individuals closer to the age of college entry.

I also find suggestive evidence on downstream outcomes. For the 18-at-establishment cohort, college graduation rises by 2.4 percentage points once enough time has passed to complete a three-year degree. Among women in this cohort, the probability of being married falls by 3.5 percentage points. Although neither estimate is statistically significant, both point in the direction expected if expanded local access to higher education shapes subsequent educational and life-course outcomes.

This paper contributes to four strands of the literature. First, it shifts attention within the study of higher education in India from elite institutions to local, non-elite public colleges. Recent work examines the expansion of elite institutions such as the Indian Institutes of Technology (IITs) and other highly selective colleges (Sekhri, 2020; Jagnani and Khanna, 2020). Yet approximately 94 percent of students enrolled in government-funded or government-controlled institutions attend state higher education institutions, making non-elite public colleges central to the higher education system but largely absent from the evidence on it. The most closely related paper is Sekhri et al. (2024), who study an earlier phase of the policy and find particularly large effects for women using long-run cohort trends in the National Family Health Survey. This paper differs in several ways. It covers the more recent RUSA expansion, uses a staggered difference-in-differences approach that accounts for differences in when colleges open, measures treatment using verified opening dates rather than approval dates, examines differences across castes, and focuses on underserved districts. This focus

allows me to study who benefits from the program within the districts it targets.

Second, the paper speaks to a general question in the study of place-based policy: whether a program targeted at places reaches the disadvantaged people the geographic criterion was chosen to proxy for, or whether its benefits are subject to elite capture by better-off residents of those same places (Galasso and Ravallion, 2005; Niehaus et al., 2013). Educationally Backward Districts are genuinely underserved, but they are not uniformly disadvantaged, and the students best positioned to take up a new college seat are not necessarily the ones the program was designed for. The caste results in this paper provide direct evidence against elite capture, as the gains are concentrated among SC/ST and OBC students.

Third, the paper contributes to the literature on caste and higher education in India, which has focused largely on reservation policy (Bertrand et al., 2010; Bagde et al., 2016; Khanna, 2020; Frisancho and Krishna, 2016). Reservation holds the number of seats fixed and changes who occupies them. Whether adding seats close to where students live brings those from historically disadvantaged groups into higher education has received less attention, and this is what the program studied here does.

Fourth, the paper contributes to the literature on distance as a barrier to schooling. Most of this work studies interventions that lower the cost of reaching an institution that already exists, and focuses on primary and secondary schooling (Muralidharan and Prakash, 2017; Borker et al., 2021; Adukia et al., 2020). Studies of school construction, such as Duflo (2001), likewise focus on primary schooling. Building new institutions may matter even more for higher education, where relocation costs are larger and enrollment decisions are discretionary rather than near-universal.

The remainder of the paper is organized as follows. Section 2 provides background on higher education in India and the RUSA Model Degree Colleges program. Section 3 describes the data and sample construction. Section 4 presents the empirical strategy. Section 5 discusses the conceptual framework. Section 6 presents the main results, and Section 7 presents robustness checks and Section 8 concludes.

2 Background

2.1 Indian Higher Education

India has experienced substantial growth in higher education over the last two decades. Today, India has the third-largest higher education system in the world in terms of volume,[2] with approximately 1,168 universities and 45,473 affiliated colleges (AISHE, 2022). During this period, total enrollment increased nearly fourfold, reflecting a major expansion of the sector. Despite this growth, participation remains low, with a Gross enrollment Ratio (GER) 28.4% in 2021–22, compared to many middle-income countries. Furthermore, access to higher education continues to vary considerably across regions and social groups.

Figure 1 shows substantial variation in state-level GER in 2021–22. Similar disparities were already evident in 2012–13 and have remain despite the overall expansion of the sector. For example, Bihar's GER increased from 13.1% in 2012–13 to 17.1% in 2021–22, while Kerala's GER rose from 22.1% to 41.3% over the same period.

The enrollment gaps persisted not only across states but also across historically disadvantaged social groups. In 2021–22, Scheduled Caste (SC) students recorded a GER of 25.9%, while Scheduled Tribe (ST) students recorded a GER of 21.2%, compared to the national average of 28.4%. The gap was particularly pronounced among ST populations. For example in the tribal-dominated states, ST GER stood at only 13.1% in Chhattisgarh, 13.9% in Jharkhand, and 15.5% in Odisha.

One potential explanation for the observed differences in higher education participation is the unequal distribution of higher education infrastructure across regions. In 2012–13, India had a national average of 25 colleges per 100,000 population aged 18–23, but this ranged from only 6 colleges per 100,000 in Bihar and 7 in Jharkhand to 44 in Andhra Pradesh and Karnataka. The average has now increased to 30 colleges per 100,000 population; however, the state level variation remains similar. The labels in Figure 1 report the corresponding

[2]India is behind only the United States and China.

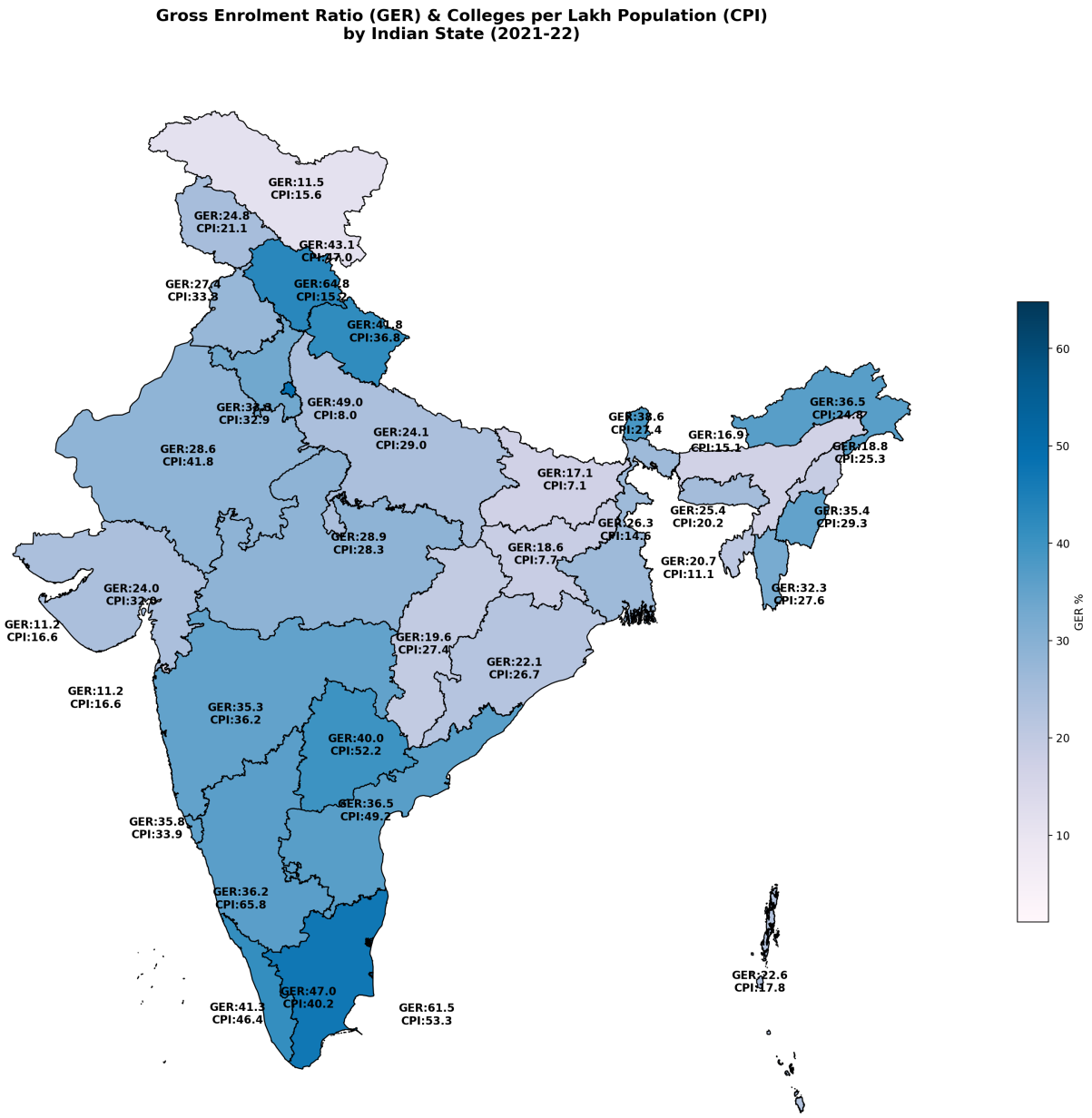


Figure 1: Gross Enrollment Ratio and College Availability by State, 2021–22

Notes: The map shows the Gross enrollment Ratio (GER) in higher education by state, with darker shading indicating higher GER. Labels report each state's GER and College Population Index (CPI). GER is total enrollment in higher education as a percentage of the population aged 18–23. CPI is the number of colleges per 100,000 population aged 18–23. Source: AISHE (2022).

College Population Index (CPI) values for each state. District-level variation is even more pronounced: 139 districts had fewer than 10 colleges, while only 40 districts had more than 200 colleges. To examine this relationship more directly, Figure 2 plots state-level GER against CPI and reveals a positive association between the two measures. These patterns suggest that disparities in higher education participation may partly reflect unequal access to higher education infrastructure.

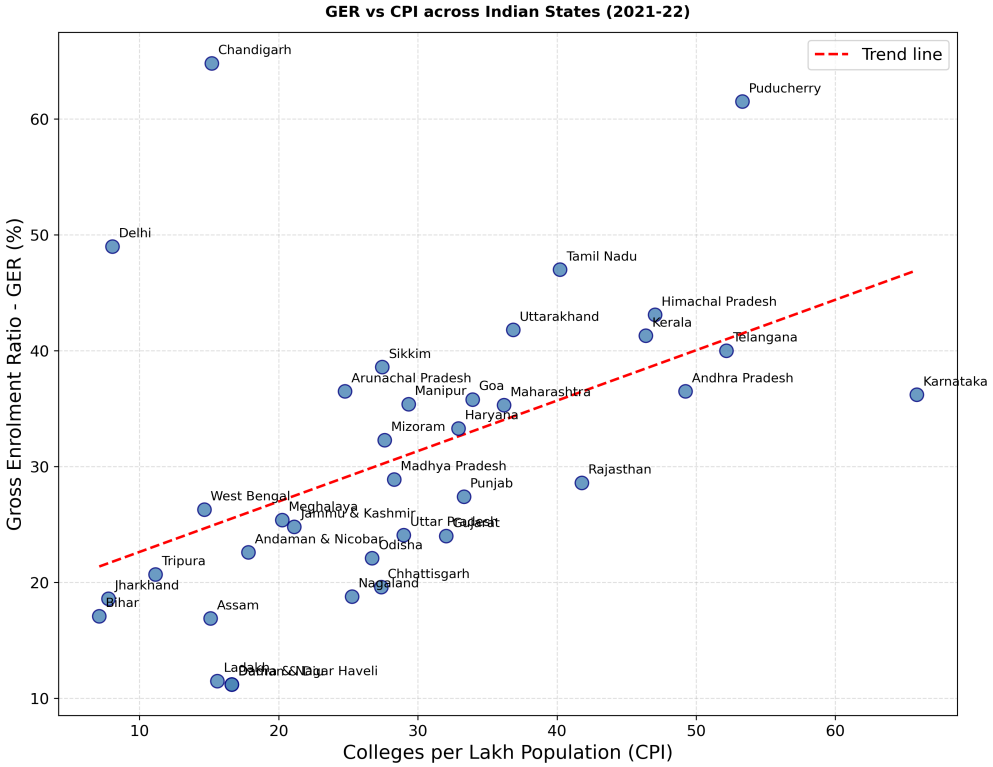


Figure 2: College Availability and Gross Enrollment Ratio across States, 2021–22

Notes: Each point represents an Indian state. The horizontal axis shows the College Population Index (CPI), the number of colleges per 100,000 population aged 18–23. The vertical axis shows the Gross Enrollment Ratio (GER), total enrollment in higher education as a percentage of the population aged 18–23. The dashed line is a linear fit. Source: AISHE (2022).

The uneven expansion of higher education across regions may partly stem from the private-sector-led nature of higher education growth in India. Over the past two decades, much of the growth in the sector has been driven by the rapid proliferation of private colleges and deemed universities. While this expansion increased the overall capacity of the higher education system, private institutions tended to concentrate in areas with stronger economic

conditions and greater demand. Consequently, many educationally disadvantaged regions remained dependent on public institutions for access to higher education.

Government colleges play a central role in India's higher education system, accounting for nearly one-third of total enrollment despite constituting only around 21% of institutions in 2021–22. While they remain important across the country, their importance is particularly pronounced in underserved regions where private provision remains limited. Colleges in low-access states are also heavily overcrowded, with average enrollment per college reaching 2,018 in Bihar and 1,934 in Jharkhand, compared to the national average of 715. These patterns suggest that low higher education participation in many districts reflects constraints in the local supply of colleges rather than low demand. Such supply-side disparities motivated large-scale public investment policies aimed at expanding higher education infrastructure in underserved districts, most notably through the Rashtriya Uchchatar Shiksha Abhiyan (RUSA).

2.2 Rashtriya Uchchatar Shiksha Abhiyan (RUSA)

The Rashtriya Uchchatar Shiksha Abhiyan (RUSA) is a centrally sponsored higher education program launched by the Government of India in 2013 to improve equity, access, and quality in public higher education. The program was introduced to expand higher education opportunities, particularly in underserved and educationally disadvantaged regions, reduce regional disparities, and improve access for marginalized groups. It also seeks to strengthen public institutions by improving infrastructure and creating more opportunities for students to enroll in higher education.

Under RUSA, the central government provides funding to states, which then allocate resources to higher education institutions based on their development plans. States are required to submit State Higher Education Plans (SHEPs) outlining their strategies for improving access and educational quality.

Within this umbrella scheme, several key components aim to expand and improve public

higher education institutions. These include establishing new universities, upgrading autonomous colleges to universities, and creating cluster universities to expand institutional capacity. The scheme also supports establishing new model colleges, upgrading existing degree colleges to model colleges, and providing infrastructure grants to universities and colleges to improve physical and academic resources.[3]

These interventions are particularly important given the structure of the higher education system in India, where around 94% of students in government-funded or government-controlled institutions are enrolled in the state higher education system, while centrally funded institutions account for a relatively small share of total enrollment despite receiving substantial resources. This concentration highlights the central role of state higher education institutions in expanding access and improving outcomes.

2.3 Model Degree Colleges

This study focuses on the Model Degree Colleges (MDCs) component of RUSA, which supports the establishment of new colleges in educationally backward districts (EBDs) identified by low GER in higher education. EBDs were defined as districts with a GER below the national average of 12.5 percent based on the 2001 Census. Under the scheme, 60 new Model Degree Colleges were to be established in these districts, with funding allocations determined by the Project Approval Board based on recommendations submitted through the State Higher Education Plans (SHEPs).

Funding priority was assigned by grading EBDs into Category A and Category B. Districts with 0–1 colleges were classified as Category A and given the highest priority, while districts with 2–4 colleges were classified as Category B. Within Category B, priority was determined based on additional criteria, including a low College Population Index (CPI),

[3]In addition, RUSA promotes research, innovation, and quality improvement, as well as equity initiatives to increase participation among disadvantaged groups. Other components include faculty recruitment support, faculty improvement programs, vocationalization of higher education, leadership development of educational administrators, and institutional restructuring and reforms. The scheme also strengthens polytechnics and supports management, monitoring, evaluation, and data systems to improve accountability and institutional performance.

defined as the number of colleges per 100,000 individuals in the 18–23 age group, a high proportion of socially and economically disadvantaged populations, low female GER, and low GER among Scheduled Caste and Scheduled Tribe populations.

States proposed the eligible districts, and the Project Approval Board reviewed these proposals between 2013 and 2015. Across ten approval meetings, states proposed approximately 140 colleges,[4] of which 103 received funding support.[5] The approved colleges became operational at different points between 2015 and 2023, resulting in a staggered expansion of higher education infrastructure across Educationally Backward Districts (EBDs). This staggered implementation generates variation across districts in the timing of access to publicly funded colleges.

To construct the treatment measure, I match approved districts to institution-level records from the All India Survey on Higher Education (AISHE) and verify actual college establishment using college names, opening years, and institutional information. The resulting treatment measure captures realized college openings rather than administrative approvals alone. The empirical analysis compares eligible EBDs that received a Model Degree College with those that did not, exploiting variation in college establishment and its timing under a staggered rollout design. This variation is used to estimate the effects of higher education expansion on educational attainment and labor market outcomes.

3 Data

For this analysis, I use two datasets: All India Survey on Higher Education (AISHE) and Periodic Labor Force Survey (PLFS). First, AISHE provides institution-level information on higher education infrastructure across India. It covers all universities and colleges and includes details such as the year of establishment, the type of institution such as universities,

[4] One college per district was allowed.

[5] These approvals included both new Model Degree Colleges and colleges funded that were approved from previous policy round. While eligibility depended primarily on EBD status and the priority criteria described above, funding decisions also reflected budget constraints and the timing of state proposals.

degree colleges, and model colleges, and their district and state location. This information allows me to identify where and when public higher education institutions were established under the program and construct district-level measures of college expansion.

Second, I use the annual rounds of the Periodic Labour Force Survey from 2017 to 2023, which is a cross-sectional household survey, that provides detailed labor market and demographic information at the individual level. From this dataset, I construct outcomes related to educational attainment, graduation rate and employment status.

3.1 Sample Construction

To identify treatment, I combine Project Approval Board records with institution-level AISHE data. I manually matched the approved districts to colleges using institution names, location information, and establishment years. The treatment date is defined as the year in which the Model Degree College became operational rather than the year of administrative approval. Since colleges opened at different points between 2015 and 2023, districts enter treatment at different times, generating a staggered adoption design.

To improve comparability between treated and control districts, I further restrict the sample to districts that experienced limited higher education expansion outside the RUSA program. Using AISHE records, I calculate the number of government colleges established in each district since 2012 and restrict the sample to districts that did not see massive expansion. The sample for both treated and control has districts that did not have more than 4 government colleges opened in the last 10 years. This restriction removes large urban districts where substantial growth in higher education infrastructure may confound the effects of the Model Degree College intervention. For example, some districts received a Model Degree College despite having relatively developed higher education systems because the targeted EBD block faced unique challenges such as conflict or geographic isolation. Raipur is one such case. Excluding districts with substantial higher education expansion helps ensure that treatment reflects the introduction of a new local college rather than

broader urban educational growth.

Finally, I restrict the sample to states containing both treated and untreated eligible EBDs. This ensures that treatment effects are identified from comparisons among districts facing similar state-level policies, economic conditions, and higher education environments, while reducing contamination of the control group from states in which all eligible districts received support.

Figure 3 shows the number of Model Degree Colleges established between 2015 and 2020, highlighting the staggered rollout of the program over time. For this analysis, I focus on districts with year of establishment from 2018–2020 as PLFS is only available from 2017 onwards.

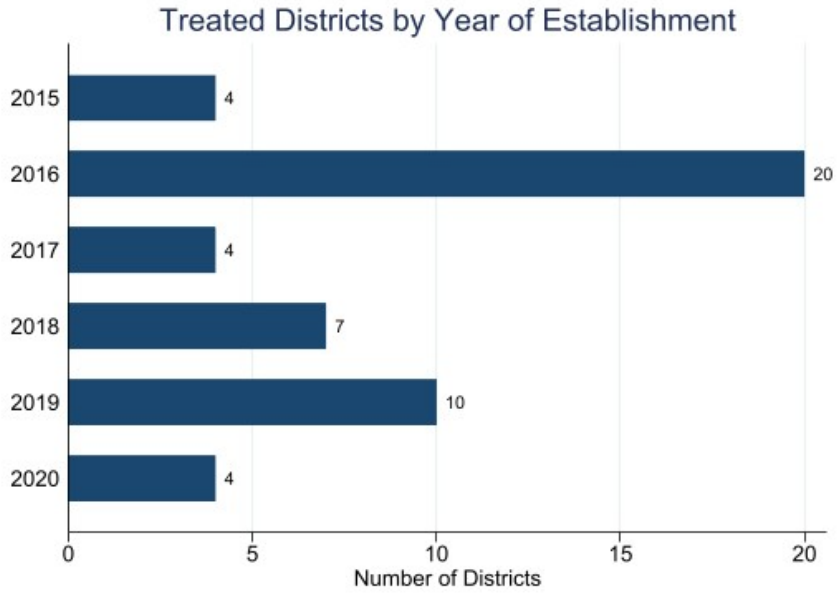


Figure 3: Number of Districts Treated by Year

Notes: RUSA Model Degree Colleges were established across six states — Assam, Bihar, Chhattisgarh, Madhya Pradesh, Odisha, and Uttar Pradesh — between 2015 and 2020 in recommended Educationally Backward Districts. The analysis focuses on the 2018–2020 establishment cohorts, which comprise 21 treated districts. Colleges established in 2015–2017 are excluded from the treated group as they predate the PLFS sample window and do not allow for pre-treatment identification. Never-treated recommended EBD districts across all six states serve as the control group.

The analysis focuses on cohorts most likely to be directly affected by the expansion of local higher education access. Figure 4 plots the age profile of school and college participation

using national sample of the data. The figure shows the share of individuals at each age who report being a student and the share who report being enrolled in college. School participation declines with age, while college enrollment rises sharply during the late teenage years and peaks around age 18–19. At age 19, both series stabilize, indicating that most transitions into college occur by the age 19. This pattern motivates the focus on individuals aged 18 at the time of college entry.

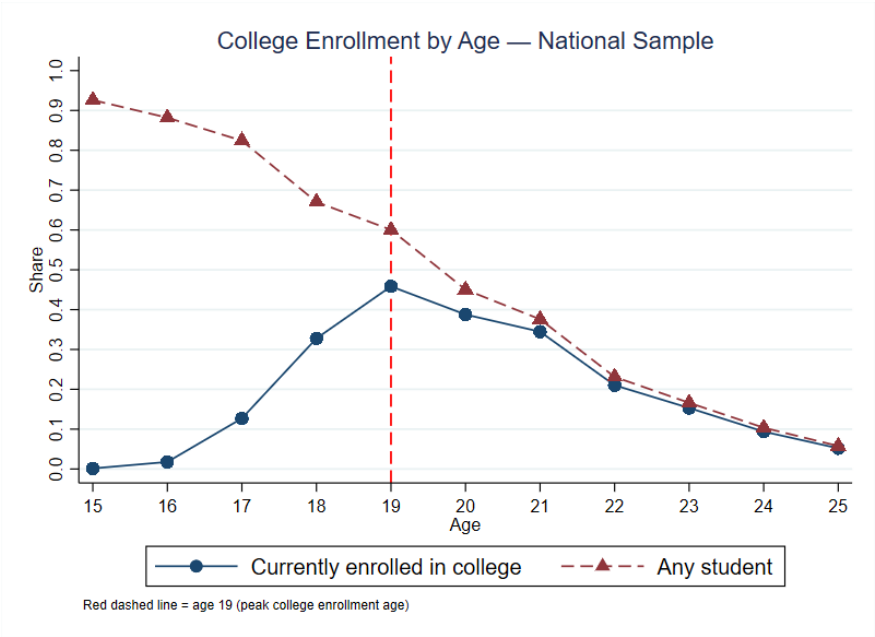


Figure 4: Share of students in college by age

Notes: The figure plots the share of individuals at each age who report being a student as their principal activity code (red line) and the share currently enrolled in college or above (blue line), using the full national PLFS sample pooled across survey rounds 2017–2023. Student status captures any educational participation including school, while college enrollment captures participation specifically at the graduate level or above. This motivates 18 at establishment cohort for main analysis.

I examine individuals who were 18 years old at the time a Model Degree College became operational in their district. These cohorts are at the margin of college-entry decisions and therefore most likely to respond to reductions in the cost of accessing higher education.

Because the PLFS begins in 2017, the main analysis initially focuses on colleges established between 2018 and 2020, for which at least one pre-treatment survey round is observed for the relevant cohorts. This allows the estimation to rely on genuine pre-treatment obser-

vations. Later, I extend the analysis to earlier establishment cohorts using an imputation approach to address missing pre-treatment observations.

3.2 Summary Statistics

Table 1 shows the baseline characteristics of these districts for individuals aged 17–19 in the year 2017, which serves as the pre-treatment period for all establishment years (2018-2020) considered in the analysis. The summary statistics show that treated and never-treated districts are similar at baseline. In the sample, 63.7% of individuals in treated districts are students compared with 62.6% in control districts, while college enrollment is 20.2% and 20.9%, respectively. The shares employed or engaged in domestic work are also similar across the two groups. The patterns are similar for men and women as well. The only notable difference is that women in treated districts are slightly less likely to be married (9.2% versus 12.3%). Overall, the differences are small, suggesting that treated and control districts are well balanced before the colleges open.

4 Empirical Strategy

For this analysis, I exploit the variation in the timing of establishment of RUSA-funded Model Degree Colleges across Educationally Backward Districts (EBDs). The treatment group consists of EBDs that received and established a Model Degree College under RUSA, while the control group consists of EBDs that satisfied the program's eligibility criteria but did not receive a college.[6] Because all districts in the sample were classified as educationally backward and considered eligible for support, this comparison helps isolate the effects of college expansion from broader differences between educationally advanced and disadvantaged regions.

[6]These districts did not receive a college because of the program's cap of 60 new colleges or limited state implementation capacity. Using the states' Detailed Project Reports, I confirm that states followed the GER requirement and proposed the districts with the lowest GER for new colleges.

Table 1: Baseline Characteristics of Treated and Never-Treated Districts — Ages 17–19

	Treated	Control	Difference	p-value
Panel A: Overall (N = 923 treated, 2,514 control)				
Student	0.637	0.626	+0.011	0.570
Enrolled in college	0.202	0.209	−0.008	0.621
Employed	0.114	0.119	−0.005	0.676
Domestic work	0.179	0.192	−0.013	0.375
Married	0.049	0.061	−0.012	0.177
Female share	0.447	0.456	−0.009	0.647
Panel B: Males (N = 510 treated, 1,367 control)				
Student	0.692	0.691	+0.001	0.971
Enrolled in college	0.216	0.211	+0.005	0.841
Domestic work	0.008	0.017	−0.009	0.146
Panel C: Females (N = 413 treated, 1,147 control)				
Student	0.569	0.549	+0.020	0.489
Enrolled in college	0.184	0.207	−0.023	0.326
Domestic work	0.390	0.401	−0.011	0.690
Married	0.092	0.123	−0.031	0.091*

Notes: Baseline activity status in 2017 for individuals aged 17–19 in treated districts (establishment years 2018–2020) and never-treated EBD districts across six states. Student: principal activity is attending any educational institution (PLFS code 91). Enrolled: currently attending college or above (attendance codes 28–32). Domestic work: principal activity is domestic duties (codes 92–93). p-values from two-sample t-tests with equal variances. No variable differs significantly at the 5% level between treated and control districts.
* $p < 0.10$.

Furthermore, as discussed in the previous section, RUSA includes multiple intervention components, most of which focus on improving the quality of existing institutions rather than creating new ones. In contrast, the Model Degree Colleges (MDCs) component is the only intervention that involves the establishment of new colleges across districts. Since other components are implemented more uniformly across eligible districts, the staggered rollout of MDCs provides the primary source of cross-district variation used in this study. This feature allows the analysis to isolate the effects of new college establishment from broader quality-improving interventions under RUSA.

4.1 Gardner (2022)

I estimate the effects of local college expansion on educational attainment and labor market outcomes using the two-stage difference-in-differences estimator proposed by Gardner (2022) . The analysis exploits the staggered establishment of RUSA Model Colleges across Educationally Backward Districts (EBDs). This approach addresses biases from standard two-way fixed effects models under staggered treatment timing.

In this method, the estimator proceeds in two steps. In the first stage, I estimate district, survey-year, and age fixed effects using only untreated observations (never treated and not yet treated):

$$z_{i,d,s} = \alpha_d + \delta_s + \gamma_a + \epsilon_{i,d,s}$$

where α_d captures district fixed effects, δ_s captures survey-year fixed effects, and γ_a captures age fixed effects. The residual $\epsilon_{i,d,s}$ is the outcome net of district, survey-year, and age fixed effects.

In the second stage, I estimate the following event-study regression:

$$\hat{\epsilon}_{i,d,s} = \sum_{t \neq -1} \pi_t \, \mathbf{1}\{\text{EventTime}_{d,s} = t\} + u_{i,d,s}$$

where $\text{EventTime}_{d,s}$ measures years relative to the establishment of a Model Degree College in district d, and $t = -1$ (the year immediately preceding treatment) is omitted as the reference period. The coefficients π_t trace the dynamic effects of treatment before and after college establishment and allow for a test of the parallel trends assumption through the pre-treatment coefficients.

The primary outcomes of interest are years of formal education, graduation rate and marital status. I define exposure cohorts based on age at the time a Model Degree College became operational in the district. As mentioned in sample construction, the analysis focuses on individuals aged 18 at the time of establishment, corresponding to the age range in which higher education enrollment decisions are typically made in India.

The main identifying assumption is that, in the absence of a Model Degree College, outcomes for eligible cohorts in treated Educationally Backward Districts (EBDs) would have evolved similarly to outcomes for the same cohorts in eligible untreated EBDs. Identification therefore comes from comparing individuals exposed to a Model Degree College in treated districts with individuals of the same age in eligible untreated districts.

5 Conceptual Framework

When a new college opens in a district, an individual's response to that opportunity depends on two key factors: (i) their stage in the transition from school to college, and (ii) the main constraints that influence their educational choices.

Who can respond and when. As shown in Figure 4, the transition into college occurs primarily between ages 17 and 19, with most individuals are in college by age 19. The effect of a new college depends on where an individual is in the education cycle when it becomes operational. Individuals who are 17 years old are typically still in secondary school and have sufficient time to adjust their college plans in response to improved local access. Those who are 18 are at the margin of the college-entry decision, while individuals who are already 19 have completed this transition, making it less likely that those who have dropped out of education will return. Consequently, I expect the effects on educational attainment to be larger for the cohorts that are transitioning and have a runway to respond to the new college establishment.

Why the response may differ by gender. Men and women face different constraints to continuing their education, which may lead to heterogeneous responses to the opening of a local college. According to the PLFS 2023–24, nearly 70 percent of men aged 15–20 who leave education report doing so to support household income, compared with about 20 percent of women. A nearby college reduces the direct costs of accessing higher education, but it may not substantially reduce the opportunity cost of remaining in school. Therefore,

for men, improved access may increase college participation by lowering geographic barriers, although financial pressures and the need to contribute to household income may continue to limit educational persistence.

Whereas for women, geographic access is likely to be a particularly important constraint. Nearly 60 percent of women aged 15–20 who leave education report domestic responsibilities as the primary reason. Moreover, previous studies show that safety concerns and mobility restrictions reduce women's educational attainment when educational institutions are located farther from home.(Jayachandran, 2015; Borker et al., 2021; Muralidharan and Prakash, 2017) Next, for women marriage decisions also overlap closely with the college-entry period,[7] further limiting educational opportunities in a patrilocal society. At the same time, previous studies show that education can carry important returns in the marriage market for women, creating incentives to invest in higher education.(Sekhri et al., 2024) Thus, the relationship between marriage and education may operate through two opposing channels: early marriage and family responsibilities can constrain educational attainment, while the potential marriage market returns to education may encourage women who enroll to persist and complete their degrees.

Overall, the opening of a local college may affect men and women differently because it interacts with the distinct constraints they face, with access barriers playing a potentially larger role for women and financial constraints playing a potentially larger role for men.

[7]The legal age of marriage is 18 for women and 21 for men

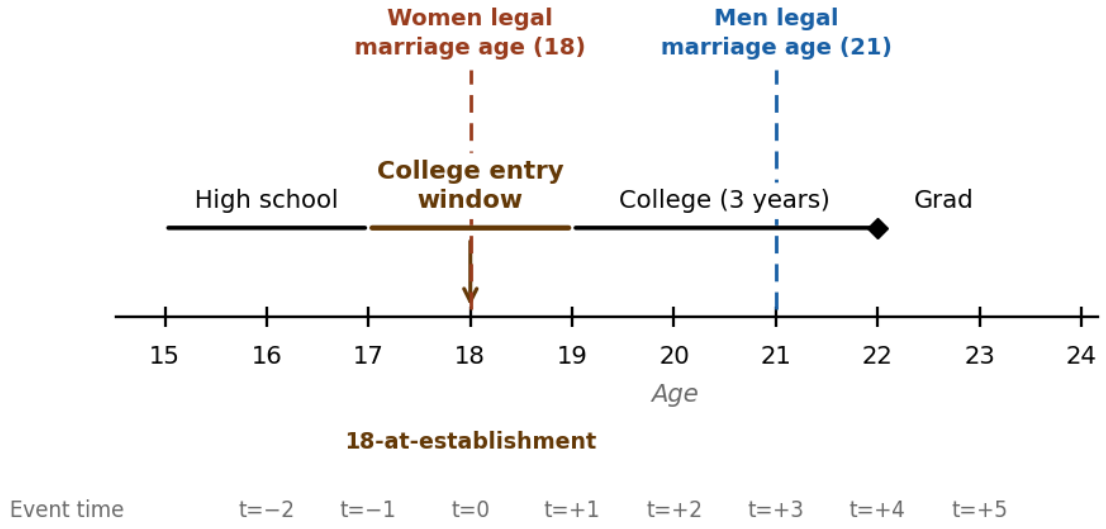


Figure 5: The figure illustrates the expected timing of treatment effects across outcomes, with event time centered on individuals who are 18 years old at establishment ($t = 0$)

Why the response may differ by social group Along with the different response by gender, the educational responses to college establishment may differ across social groups because social norms and other constraints may affect them differently. For historically disadvantaged groups (SC/ST), the perceived returns to education, norms around mobility, and safety may make access to a nearby college especially important — or may limit the benefits of greater access if these norms continue to discourage college attendance.

What this implies for outcomes over time. Figure 5 illustrates the expected timing of the treatment effects relative to the establishment of a Model Degree College. Event time is centered on individuals who are 18 years old when the college is established ($t = 0$), corresponding to the typical age of college entry. If local college access increases enrollment, years of schooling should begin to rise from $t = 0$ onward. Graduation effects should appear around $t = +3$ to $t = +5$, reflecting the typical three-year duration of a bachelor's degree in India. Delayed marriage is also expected to emerge for the cohorts experiencing schooling gains, as students who remain enrolled are more likely to postpone marriage. Where as, employment effects are less likely to be observed within our sample period because treated

individuals are still completing their degrees, and most would not have entered the labor market by the end of our observation window.

6 Results

This section presents the effects of RUSA Model Degree Colleges on educational attainment and labor market outcomes using the staggered difference-in-differences design. The main analysis focuses on individuals who were 18 years old when the college was established, consistent with the primary cohort specified in the conceptual framework. I extend the analysis to younger cohort that is 17 years old at establishment. I report the results for the 19-year-old cohort as placebo test with other robustness, along with additional analyses using the imputed sample, in the Appendix.

6.1 Years of Formal Education

The first outcome is years of formal education. As discussed in the framework, I expect the first and largest response to the opening of a college among individuals who were 18 years old at establishment, as this is the age group most likely to respond by continuing their education. Figure 6 presents the event-study estimates for years of formal education.

The results show that opening a RUSA Model Degree College increased educational attainment in treated districts relative to eligible districts that never received a college. The event-study estimates show positive effects around $t = 0$, $+1$, and $+4$, which correspond roughly to key points in the schooling progression. These patterns suggest that the opening of a college allowed individuals to continue their education at important transition points.

Table 2 reports the pooled ATT for individuals who were 18 years old when the college was established. Opening a RUSA Model Degree College increases years of formal education by 0.632 years for this group.

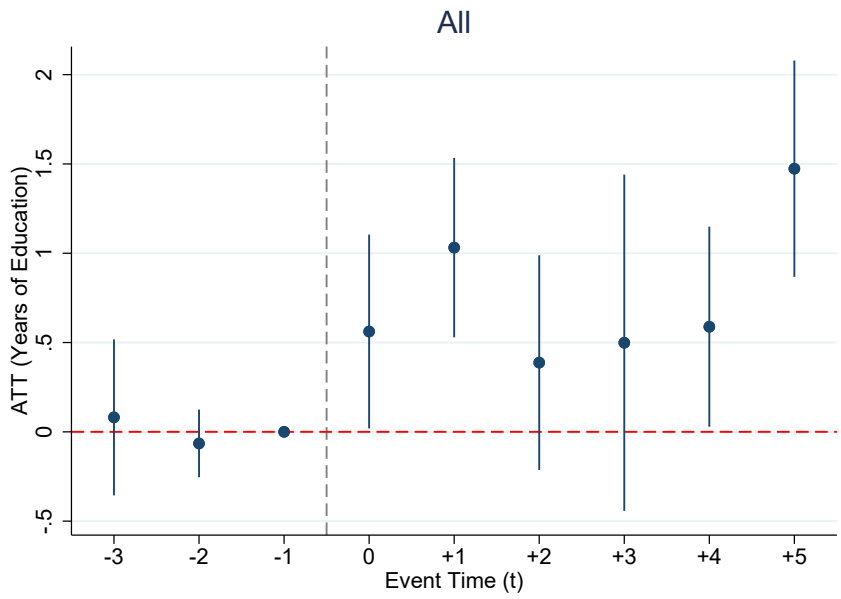


Figure 6: Effect on Years of Formal Education

Notes: The figure presents event-study coefficients from the two-stage DiD estimator for individuals who were 18 years old when the college was established. The analysis uses colleges established between 2018–2020, with the affected birth cohorts 2000–2002. The dashed vertical line at $t = -0.5$ separates pre- from post-treatment periods. The balanced event window is $t \in [-1, +3]$. Estimates at $t = +5$ are identified from the 2018 establishment-year cohorts only. Whiskers represent 95% confidence intervals. Standard errors are clustered at the district level.

6.1.1 Heterogeneous Effects by Gender

The gender-specific estimates show that the effect is primarily driven by men. Among individuals who were 18 years old when the college was established, men gain 0.925 years of formal education, while the estimate for women is positive but not statistically significant. This suggests that the increase in educational attainment following the opening of a RUSA Model Degree College is concentrated among men.

Figure 7 presents the corresponding event-study estimates by gender for individuals who were 18 years old when the college was established. The estimates show a clear increase in years of formal education for men following the college opening, while the estimates for women remain close to zero. The figure therefore supports the pattern in Table 2, with the

Table 2: Effect on Years of Formal Education by Gender

	All	Males	Females
Panel A: 18-at-Establishment			
ATT	0.632***	0.925***	0.357
SE	(0.218)	(0.301)	(0.312)
N	17,131	8,998	8,133
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Age FE	Yes	Yes	Yes

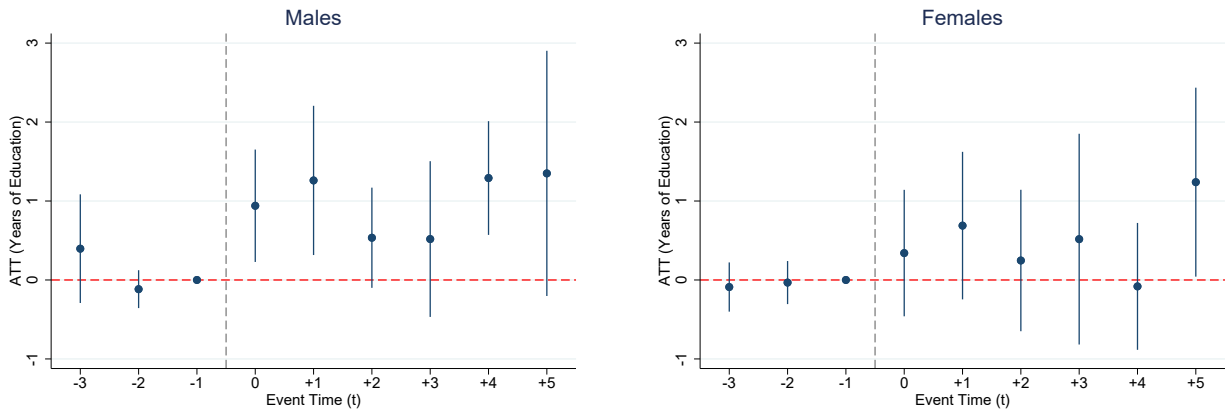
Notes: Each cell reports the ATT from a separate regression using the two-stage DiD estimator with exact matching of the birth cohort for the control group. All specifications include age, district, and year fixed effects in the first stage. The analysis focuses on individuals who were 18 years old when the college was established. Results are robust across control group definitions, estimators, and sample restrictions (Table 4). Standard errors clustered at the district level are reported in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

increase in educational attainment primarily driven by men.

6.1.2 Heterogeneous Effects by Social Group

Next, I examine whether the effect on years of formal education differs across social groups. Table 3 shows that the largest effects among individuals who were 18 years old when the college was established occur for SC/ST and OBC individuals. Years of formal education increase by 0.881 years for SC/ST individuals and 0.835 years for OBC individuals, while the estimated effect for General-category individuals is smaller and statistically insignificant at 0.323 years. Within the SC/ST and OBC groups, the gains are concentrated among males, while the female estimates are not statistically significant.

Figure 8 presents the corresponding event-study estimates by social group. The larger effects for SC/ST and OBC individuals suggest that the expansion of local college access particularly benefited historically disadvantaged groups. This pattern is consistent with the



(a) Males (b) Females

Figure 7: Effect on Years of Formal Education by Gender

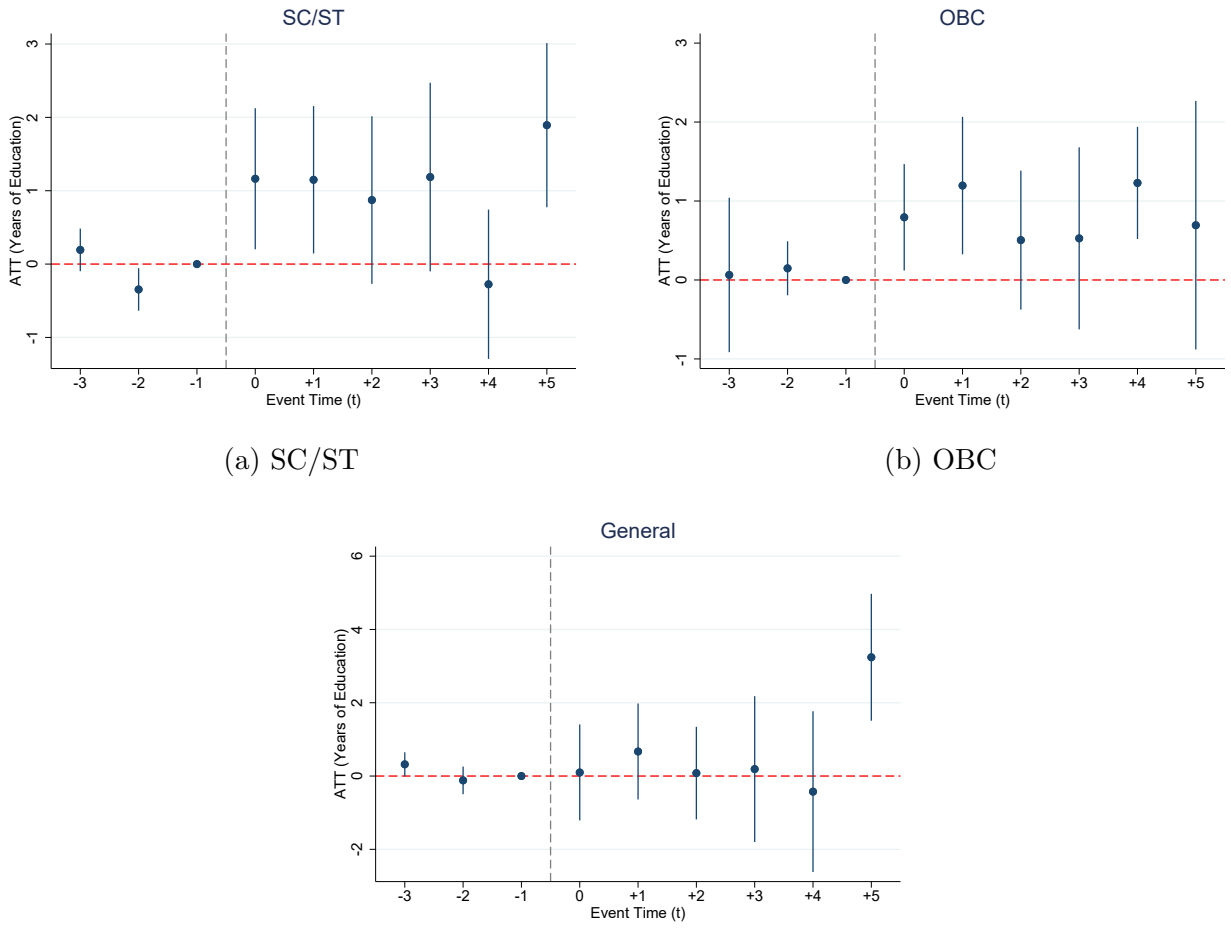
Notes: The figure presents event-study coefficients from the two-stage DiD estimator for individuals who were 18 years old when the college was established. The analysis uses colleges established between 2018–2020, with affected birth cohorts 2000–2002. The dashed vertical line at $t = -0.5$ separates pre- from post-treatment periods. The balanced event window is $t \in [-1, +3]$. Estimates at $t = +5$ are identified from the 2018 establishment-year cohorts only. Whiskers represent 95% confidence intervals. Standard errors are clustered at the district level.

objective of RUSA, which targeted Educationally Backward Districts with limited access to higher education.

Table 3: Heterogeneity by Social Group: Effect on Years of Formal Education

	SC/ST			OBC			General		
	All	Males	Females	All	Males	Females	All	Males	Females
ATT	0.881**	0.977**	0.596	0.835***	1.118**	0.217	0.323	0.093	0.897
SE	(0.345)	(0.482)	(0.475)	(0.293)	(0.491)	(0.540)	(0.503)	(0.600)	(0.775)
N	5,011	2,564	2,447	9,003	4,788	4,215	3,117	1,646	1,471
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Age FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Each cell reports the ATT from a separate regression using the two-stage DiD estimator. The sample includes individuals who were 18 years old when the college was established in recommended EBD districts across six states, using 2018–2020 establishment cohorts. The control group is selected by exact matching on birth cohort. SC/ST: Scheduled Caste and Scheduled Tribe. OBC: Other Backward Class. General: unreserved category. Standard errors are clustered at the district level.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.



(a) SC/ST

(b) OBC

(c) General

Figure 8: Effect on Years of Formal Education by Social Group

Notes: The figure presents event-study coefficients from the two-stage DiD estimator for individuals who were 18 years old when the college was established. The analysis uses exact-match control cohorts and includes age fixed effects in the first stage. SC/ST: Scheduled Caste and Scheduled Tribe. OBC: Other Backward Class. General: unreserved category. The dashed vertical line at $t = -0.5$ separates pre- from post-treatment periods. The balanced event window is $t \in [-1, +3]$. Whiskers represent 95% confidence intervals. Standard errors are clustered at the district level.

6.1.3 Robustness to different specifications

I next examine the robustness of the estimates to alternative specifications. I consider wider control cohort windows in addition to the exact-match specification. Expanding the control group includes individuals who were older at the time of college establishment. With age fixed effects, these additional observations help estimate the age profile more precisely, which is important because the treated cohort becomes one year older with each event time. I also

consider a fixed effects specification, state-by-year fixed effects, and a balanced event-study window that I observe all my year of establishment even times.

Table 4 presents the robustness results for the years of formal education. The estimates remain stable across specifications, with the overall effect ranging from 0.464 to 0.614 years, compared with 0.632 years in the baseline specification. The effect remains statistically significant across specifications. The estimates are also consistently larger for men, ranging from 0.560 to 0.931 years, while the female estimates remain positive but statistically insignificant. Overall, the results are robust to alternative control groups, fixed effects specifications, and event-study windows.

Table 4: Robustness: Effect on Years of Formal Education — 18-at-Establishment

Row	Specification	All	Males	Females
1	Baseline: Gardner, exact match, Age FE *(main)*	0.632***	0.925***	0.357
		(0.218)	(0.301)	(0.312)
2	Mid control cohort window, Age FE	0.614***	0.931***	0.314
		(0.217)	(0.305)	(0.314)
3	Wide control cohort window, Age FE	0.586***	0.903***	0.300
		(0.216)	(0.299)	(0.313)
4	TWFE with Age FE, exact match	0.516**	0.560**	0.405
		(0.221)	(0.253)	(0.320)
5	Baseline + State×Year FE	0.464**	0.814**	0.157
		(0.219)	(0.317)	(0.305)
6	Balanced window $t = -1$ to $t = +3$	0.599**	0.829**	0.309
		(0.287)	(0.389)	(0.326)
Observations (exact match)		17,131	8,998	8,133
District FE		Yes	Yes	Yes
Year FE		Yes	Yes	Yes
Age FE		Yes	Yes	Yes

Notes: Outcome: years of formal education. Sample: individuals aged 18 at college establishment in recommended EBD districts across six states (2018–2020 establishment cohorts). Exact match control: birth cohorts 2000–2002. Mid control: birth cohorts 1998–2002. Wide control: birth cohorts 1996–2002. Gardner (2022) two-stage DiD unless noted. TWFE estimated using `areg` with absorbed district FE. Balanced window restricts to $t = -1$ to $t = +3$ where all three establishment cohorts contribute observations. Standard errors clustered at district level.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6.1.4 Mechanisms: Educational Trajectories and Gender Differences

The main results show that opening a RUSA Model Degree College increases years of formal education by 0.632 years for individuals who were 18 when the college was established, with the effect concentrated among men. Men gain 0.925 years of formal education, while the estimate for women is positive but not statistically significant. I next examine changes in educational attendance and reported reasons for leaving education to understand the source of this gender difference in the post event periods.

The activity patterns in Table 5 show that both men and women initially respond to the opening of a local college by increasing college enrollment. Among men, enrollment rises from 5.1 percent at $t = -1$ to 22.2 percent at $t = 0$ and peaks at 43.7 percent at $t = +1$. Among women, enrollment increases from 14.3 percent at $t = -1$ to 20.1 percent at $t = 0$ and 36.5 percent at $t = +1$. After this initial increase, however, the trajectories diverge. For men, the decline in enrollment is accompanied by a substantial increase in employment, which rises from 23.7 percent at $t = +1$ to 57.1 percent at $t = +3$ and approximately 64 percent at $t = +4$ and $t = +5$. For women, employment increases much less, while domestic work and marriage rise sharply. By $t = +4$, 76.9 percent of women report domestic work and 52.1 percent are married.

The second set of statistics provides further insight into these different educational trajectories by examining reported reasons for not attending education. Among men, college attendance peaks at 43.7 percent at $t = +1$ and subsequently declines, while the share reporting that they are not attending education to supplement household income rises from 14.8 percent at $t = +1$ to 34.1 percent at $t = +2$, 50.0 percent at $t = +3$, and 68.0 percent at $t = +5$. Among women, college attendance similarly declines after $t = +1$, but domestic responsibilities become increasingly prominent as a reported reason for non-attendance, rising to 41.1 percent at $t = +2$ and 69.7 percent at $t = +5$. Taken together, these descriptive patterns suggest that both men and women initially respond to the availability of a local college, but face different constraints in remaining in education: men increasingly leave ed-

ucation to supplement household income and enter employment, while women increasingly leave education because of domestic responsibilities and marriage.

These patterns are consistent with the conceptual framework and suggest that reducing the geographic cost of college access increases participation and years of schooling, particularly among men. However, the descriptive patterns also indicate that income constraints remain binding. As men progress beyond the initial increase in college enrollment, many leave education to supplement household income and enter employment. Thus, while improved local access induces some individuals to pursue higher education, it may not be sufficient to ensure college completion for those facing strong income constraints. These constraints may lead students to delay completion or leave college before obtaining a degree, limiting the overall effect of increased access on educational attainment.

Table 5: Activity Status by Event Time — 18-at-Establishment Treated Districts (%)

	Panel A: Males (N pre=575, N post=743)				
t	Student	Enrolled	Employed	Domestic	Married
-1	82.2	5.1	11.5	0.6	0.0
0	68.7	22.2	23.7	2.0	0.5
$+1$	72.6	43.7	23.7	0.0	1.5
$+2$	48.2	37.2	41.5	1.2	6.7
$+3$	33.7	30.6	57.1	0.0	15.3
$+4$	25.2	23.6	64.2	0.8	23.6
$+5$	24.0	24.0	64.0	0.0	32.0
	Panel B: Females (N pre=499, N post=628)				
t	Student	Enrolled	Employed	Domestic	Married
-1	71.4	14.3	2.7	25.0	0.0
0	54.1	20.1	4.4	40.3	6.9
$+1$	51.8	36.5	9.4	36.5	18.8
$+2$	37.7	30.5	6.6	54.3	34.4
$+3$	30.1	31.3	10.8	56.6	44.6
$+4$	13.7	11.1	7.7	76.9	52.1
$+5$	18.2	15.2	3.0	78.8	54.5

Notes: Activity status expressed as percentage of individuals in each group-year cell. Sample: individuals aged 18 at college establishment in recommended EBD districts (2018–2020 cohorts). $t = -1$ corresponds to age 17 (pre-treatment baseline); $t = 0$ corresponds to age 18 (year of establishment). Student: principal activity is attending any educational institution (PLFS code 91). Enrolled: currently attending college or above (codes 28–32). Employed: any employment (codes 11, 12, 21, 31, 41, 51). Domestic: principal activity is domestic duties (codes 92–93). Married: currently married. The male pattern shows enrollment peaking at $t = +1$ (43.7%) followed by a progressive transition to employment by $t = +3$ to $t = +5$ (57–64%). The female pattern shows a sharp rise in domestic work and marriage from $t = +2$ onward, reaching 76.9% domestic and 52.1% married by $t = +4$.

	Males			Females		
t	College	Income	Domestic	College	Income	Domestic
-1	5.1	4.5	0.6	14.3	0.0	16.1
0	22.2	17.7	3.0	20.1	2.5	28.9
$+1$	43.7	14.8	0.0	36.5	8.2	24.7
$+2$	37.2	34.1	2.4	30.5	2.6	41.1
$+3$	30.6	50.0	0.0	31.3	6.0	42.2
$+4$	23.6	54.5	2.4	11.1	7.7	63.2
$+5$	24.0	68.0	0.0	15.2	9.1	69.7
N (pre)		575			499	
N (post)		743			628	

Notes: The table reports the percentage of individuals reporting current status of attendance "College" refers to enrollment in higher education, "Income" refers to left due to participation in income-generating work, and "Domestic" refers to left due to domestic work.

6.2 Marriage outcomes

As discussed in the framework, for women, the decision to continue education and the decision to marry occur around the same age, making these outcomes closely related. The negative coefficient on marriage for women, 0.035, suggests that establishing a nearby college is associated with a 3.5 percentage-point reduction in the probability of being married. Although the estimate is imprecisely estimated, its direction is consistent with the possibility that greater access to higher education may delay marriage for some women by allowing them to remain in education. This may be particularly relevant in settings with patrilocal marriage norms, where women typically move to their husband's household after marriage, making marriage and continued education potentially competing choices. The result should therefore be interpreted as suggestive rather than evidence of a broad shift away from marriage toward education. For men, the estimates are close to zero.

6.3 Graduation

For graduation, I restrict the analysis to periods with event time t $\geq$ 3, since a three-year degree is the earliest point at which students who enroll following the establishment of the

Table 6: Effect on Marriage Probability — 18-at-Establishment

	All	Males	Females
18-at-Establishment			
ATT	−0.015	0.017	−0.035
SE	(0.018)	(0.017)	(0.026)
N	17,131	8,998	8,133
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Age FE	Yes	Yes	Yes

Notes: Each cell reports the ATT from a separate regression using the two-stage DiD estimator. The outcome is an indicator for currently married (Marital Status = 2). Standard errors clustered at the district level are reported in parentheses.

college could complete a degree. This provides a meaningful window in which to examine whether the expansion in college access translates into higher graduation rates. The estimates without age fixed effects are positive, but once age fixed effects are included, the effects become much smaller: the estimated effect is 2.4 percentage points for the full sample, 3.2 percentage points for men, and 1.6 percentage points for women. These estimates are relatively small and imprecisely estimated, suggesting that there is not yet enough post-establishment time to observe a clear effect on graduation. In other words, while the earlier results show that the program increases years of schooling, the available post-treatment period is too short to determine whether these additional years of schooling ultimately translate into higher graduation rates in the district.

Table 7: Effect on Graduation Rate — 18-at-Establishment Cohort

	Model 1			Model 2		
	All	Males	Females	All	Males	Females
18-at-Establishment (birth cohorts 2000–2002)						
ATT	0.052**	0.062**	0.044	0.024	0.032	0.016
SE	(0.023)	(0.026)	(0.030)	(0.019)	(0.025)	(0.028)
N	16,808	8,812	7,996	16,808	8,812	7,996
District FE	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Age FE	No	No	No	Yes	Yes	Yes

Notes: Outcome: graduation (degree or above). The post-treatment indicator is restricted to $t \geq 3$, corresponding to the earliest period in which a three-year Indian degree could be completed following enrollment. The 2020 establishment cohort is excluded because it does not have a post-treatment period with $t \geq 3$. The exact-match control group consists of birth cohorts 2000–2002. Age fixed effects are included as a robustness specification. Standard errors clustered at the district level are reported in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6.4 Employment

The results in Table 8 examine whether the establishment of a nearby college affects employment among the 18-at-establishment cohort. The estimated effect on employment is close to zero for the full sample, at -0.004, and is not statistically significant. The estimate is slightly positive for males (0.012) and negative for females (-0.039), but neither estimate is statistically significant. Overall, these results suggest that the increase in access to higher education does not lead to a significant change in employment among this cohort. This is consistent with the interpretation that the primary response to college establishment is through educational decisions rather than changes in employment.

Table 8: Effect on Employment — 18-at-Establishment

	All	Males	Females
ATT	−0.004	0.012	−0.039
SE	(0.023)	(0.041)	(0.032)
Observations	17,131	8,998	8,133
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Age FE	Yes	Yes	Yes

Notes: Each cell reports the ATT from a separate regression using the two-stage DiD estimator with exact match control (birth cohorts 2000–2002) and age fixed effects in the first stage. Outcome: any employment, defined as principal activity being wage or self-employment (PLFS codes 11, 12, 21, 31, 41, 51). Standard errors clustered at district level in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

6.5 Effect on younger cohort

To examine whether the expansion of local college access also affects individuals who are younger and still making decisions about higher education, I extend the analysis to individuals who were 17 years old when the Model Degree College was established. At this age, individuals are approaching the transition from secondary school to higher education, so the availability of a nearby college may influence whether they continue their education and how long they remain in school. The results in Table 9 show a positive effect on years of formal education. For the full sample, the estimated ATT is 0.394 years. The effect is larger and more precisely estimated for males, who experience an increase of 0.467 years in formal education. The estimated effect for females is also positive, at 0.418 years although it is not statistically significant. Thus, the results for this younger cohort are consistent with the broader finding that expanding local college access can increase educational attainment, with the most precisely estimated gains occurring among men.

Table 9: Effect on Years of Formal Education — 17-at-Establishment

	All	Males	Females
17-at-Establishment			
ATT	0.394*	0.467**	0.418
SE	(0.229)	(0.203)	(0.328)
N	17,454	9,178	8,276
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Age FE	Yes	Yes	Yes

Notes: Each cell reports the ATT from a separate regression using the two-stage DiD estimator with exact-match birth cohorts for the control group and includes age, district, and year fixed effects in the first stage. The exact-match control cohort consists of birth cohorts 2001–2003. Standard errors clustered at the district level are reported in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Heterogeneous effects on younger cohort

The heterogeneity by social group for 17 year olds suggests that the expansion of local college access mattered most for students from historically disadvantaged communities. As shown in Table 10 Columns 1 – 6, the effects for SC/ST and OBC students are consistently positive, while the estimates for the General category in Columns 7 – 9 are close to zero. The pattern is particularly pronounced among OBC students, with gains in educational attainment for both men and women, and the largest increase appearing among OBC women. Among SC/ST students, the gains are more concentrated among men. In contrast, students in the General category appear to experience little change in educational attainment following the establishment of a nearby college. This pattern is consistent with the idea that geographic access to higher education may be a more important constraint for students from disadvantaged social groups, so reducing the distance to college can have a larger effect on their decision to remain in education.

Next, I show the robustness of my results for 17 year olds in the table 11. The overall estimates remain positive across specifications, ranging from 0.232 to 0.419 years. The

Table 10: Heterogeneity by Social Group: Effect on Years of Formal Education — 17-at-Establishment

	SC/ST			OBC			General		
	All	Males	Females	All	Males	Females	All	Males	Females
17-at-Establishment									
ATT	0.352	0.733*	0.418	0.586*	0.804*	1.063**	0.020	0.055	0.093
SE	(0.382)	(0.409)	(0.657)	(0.325)	(0.414)	(0.479)	(0.367)	(0.328)	(0.496)
N	5,091	2,629	2,462	9,206	4,893	4,313	3,157	1,656	1,501
District FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Age FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: Each cell reports the ATT from a separate regression using the two-stage DiD estimator with exact-match birth cohorts for the control group and includes age, district, and year fixed effects in the first stage. The sample consists of individuals aged 17 at college establishment in recommended EBD districts across six states (2018–2020 establishment cohorts). The exact-match control cohort consists of birth cohorts 2001–2003. SC/ST: Scheduled Caste and Scheduled Tribe. OBC: Other Backward Class. General: unreserved category. Standard errors clustered at the district level are reported in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

estimates gets smaller under state-by-year fixed effects are added, but the effect remains statistically significant in the baseline, wider control-group specifications, and balanced-window specification. As in the baseline results, the effects are concentrated among men, with male estimates ranging from 0.358 to 0.508 years. The results for females are also positive but the estimates are not statistically significant.

7 Robustness Analysis

I conduct several robustness checks to assess whether the main findings are driven by the age composition of the sample or by a small number of treated districts. First, I examine whether the estimated effects extend to individuals who were already beyond the typical transition age into higher education when the college was established. I then conduct a leave-one-out analysis to assess whether the main estimate is driven by any single treated district.

Table 11: Robustness: Effect on Years of Formal Education — 17-at-Establishment

	Specification	All	Males	Females
(1)	Baseline (main specification)	0.394*	0.467**	0.418
		(0.229)	(0.203)	(0.328)
(2)	Mid control cohort window	0.419*	0.508**	0.439
		(0.230)	(0.207)	(0.325)
(3)	Wide control cohort window	0.412*	0.476**	0.452
		(0.235)	(0.216)	(0.323)
(4)	TWFE, exact match	0.232	0.358*	0.104
		(0.210)	(0.214)	(0.245)
(5)	Baseline + State×Year FE	0.260	0.383*	0.188
		(0.233)	(0.217)	(0.324)
(6)	Balanced window, $t = -1$ to $+3$	0.408*	0.505**	0.396
		(0.221)	(0.245)	(0.273)
Observations (exact match)		17,454	9,178	8,276
District FE		Yes	Yes	Yes
Year FE		Yes	Yes	Yes
Age FE		Yes	Yes	Yes

Notes: Outcome: years of formal education. Sample: individuals aged 17 at college establishment in recommended EBD districts across six states (2018–2020 establishment cohorts). The baseline uses the Gardner (2022) two-stage DiD estimator with an exact-match control group (birth cohorts 2001–2003) and age fixed effects. Mid control: birth cohorts 1999–2003. Wide control: birth cohorts 1997–2003. All specifications include age fixed effects. TWFE is estimated using `areg` with absorbed district fixed effects. The balanced window restricts to $t = -1$ to $t = +3$, where all three establishment cohorts contribute observations. Standard errors are clustered at the district level.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7.1 Effect on older cohort

As a placebo test, I examine individuals who were 19 years old at the time of college establishment. By this age, individuals would generally have already transitioned into higher education or chosen not to enroll. Therefore, the establishment of a new college is unlikely to substantially affect their educational decisions. Table 12 reports the results for the same. The estimated effect on years of formal education is 0.203 years for the full sample, 0.333 years for males, and 0.016 years for females. None of these estimates is statistically significant. The absence of statistically significant effects for the 19-at-establishment cohort suggests that the effects of college establishment are concentrated among individuals who

were younger and more likely to be at the margin of entering higher education when the college became available. This provides supporting evidence that the main results are capturing responses to improved access to higher education rather than broader changes occurring in treated districts.

Table 12: Effect on Years of Formal Education — 19-at-Establishment

	All	Males	Females
Panel A: 19-at-Establishment			
ATT	0.203	0.333	0.016
SE	(0.231)	(0.287)	(0.276)
N	16,559	8,570	7,989
District FE	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Age FE	Yes	Yes	Yes

Notes: Each cell reports the ATT from a separate regression using the two-stage DiD estimator with exact match control (birth cohorts 1999–2001) and age, district, and year fixed effects in the first stage. Results are robust across control group definitions, estimators, and sample restrictions. Standard errors clustered at district level in parentheses.
*** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

7.2 Leave One-Out District

As a second robustness check, I assess whether the main estimate is driven by any individual treated district. Figure 9 reports leave-one-out estimates obtained by sequentially excluding each of the 21 treated districts from the analysis. The estimates remain positive across the different specifications, and the confidence intervals generally overlap with the main estimate of 0.632 years. This indicates that the main result is not driven by a single treated district and is therefore robust to the exclusion of individual treated locations.

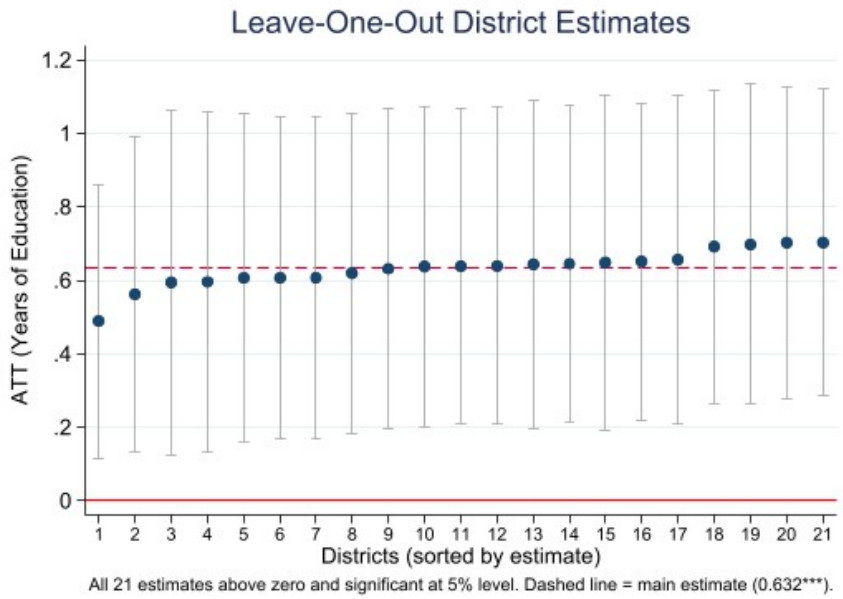


Figure 9: Leave-One-Out Robustness

Notes: Each point represents the two-stage DiD estimate of the effect of college establishment on years of formal education for the 18-at-establishment cohort, dropping one treated district at a time. The sample comprises 21 treated districts. Points are sorted in ascending order by estimate. Whiskers represent 95% confidence intervals. The dashed horizontal line marks the main estimate (ATT = 0.632, $p<0.01$) from the full sample. Exact match control: birth cohorts 2000–2002 in never-treated recommended EBD districts. Standard errors clustered at district level.

8 Conclusion

This paper evaluates the impact of improving local access to public higher education on educational attainment and labor market outcomes. I study the establishment of Model Degree Colleges under the Rashtriya Uchchatar Shiksha Abhiyan (RUSA), a centrally sponsored program targeted toward Educationally Backward Districts in India. I exploit variation in the timing of college establishment across districts and estimate the effects of a new local college on education, employment, and marriage outcomes for cohorts at the margin of college entry.

The results show that opening a local college increases educational attainment. Individuals who were 18 when a college opened gain 0.63 years of formal education, and those who

were 17 gain 0.39 years, while the effect for those who were 19 is small and not statistically significant. The gains are driven primarily by men and are concentrated among SC/ST and OBC students, which suggests that the program reached historically disadvantaged groups rather than being captured by better-off students. Women benefit less, consistent with constraints such as early marriage and domestic responsibilities that a nearby college does not remove.

I also find suggestive evidence on downstream outcomes. Graduation rises once enough time has passed to complete a degree, and the probability of marriage falls among women who were 18 at establishment, although neither estimate is statistically significant. The main findings are robust to alternative control groups, fixed effects specifications, the exclusion of individual treated districts, and the inclusion of earlier program cohorts using an imputation approach. Taken together, the results suggest that expanding the local supply of public colleges can raise educational attainment in underserved regions, but that reaching women may require policies that address constraints beyond distance.

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9 Appendix

9.1 Extending the sample using imputed observations

A limitation of the PLFS data is that surveys begin in 2017, leaving no pre-treatment observations for cohorts exposed to colleges established in 2015–2017. For these cohorts, individuals had already aged beyond the target exposure window by the time the first survey was conducted. To address this issue, the main analysis focuses on colleges established between 2018 and 2020, for which at least one pre-treatment survey round is available.

As a robustness exercise, I construct an extended sample that includes colleges established between 2015 and 2017 using a parallel cohort imputation approach. The method uses younger cohorts observed in the same district to proxy for the missing pre-treatment outcomes of the treated cohort. The main assumption is that adjacent cohorts within a district follow similar pre-treatment trends, allowing younger cohorts to serve as valid counterfactuals for the missing observations. Here , the figure shows the cohort construction with imputed rows.

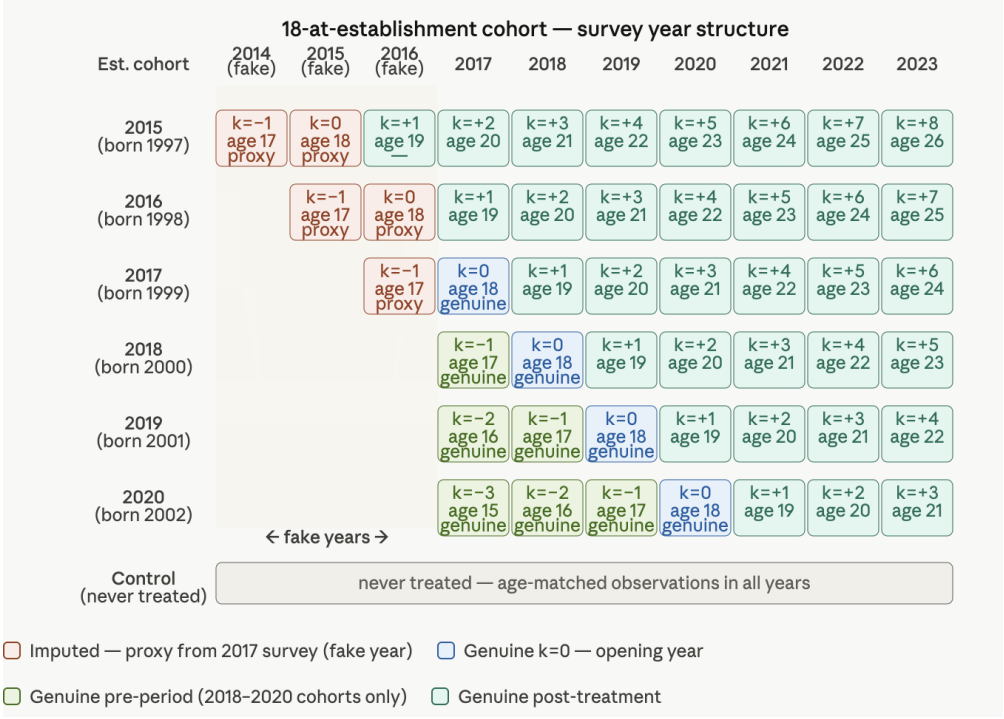


Figure 10: Sample Construction

Results for complete sample

Years of Education

Figure 10-13 presents the event-study estimates using the full imputed sample. The results closely mirror those from the baseline specification, with coefficients generally exhibiting the same signs and patterns across cohorts. However, the estimated effects are somewhat smaller in magnitude. For example, the increase in years of schooling for individuals who

were 18 years old at the time of college establishment is approximately 0.42 years, compared to the larger estimate of 0.632 years obtained in the main specification. Similar attenuation is observed for the 19-year-old cohorts.

One possible explanation for the smaller magnitudes is that the imputation procedure relies on younger cohorts as proxies for missing pre-treatment observations. If educational decisions among these cohorts were already influenced by expectations of a new college opening, the imputed pre-treatment outcomes may partially reflect treatment effects, leading to attenuated estimates. However, the similarity in the overall patterns provides reassurance that the main findings are robust to the inclusion of earlier establishment cohorts.

Age 18 at Establishment

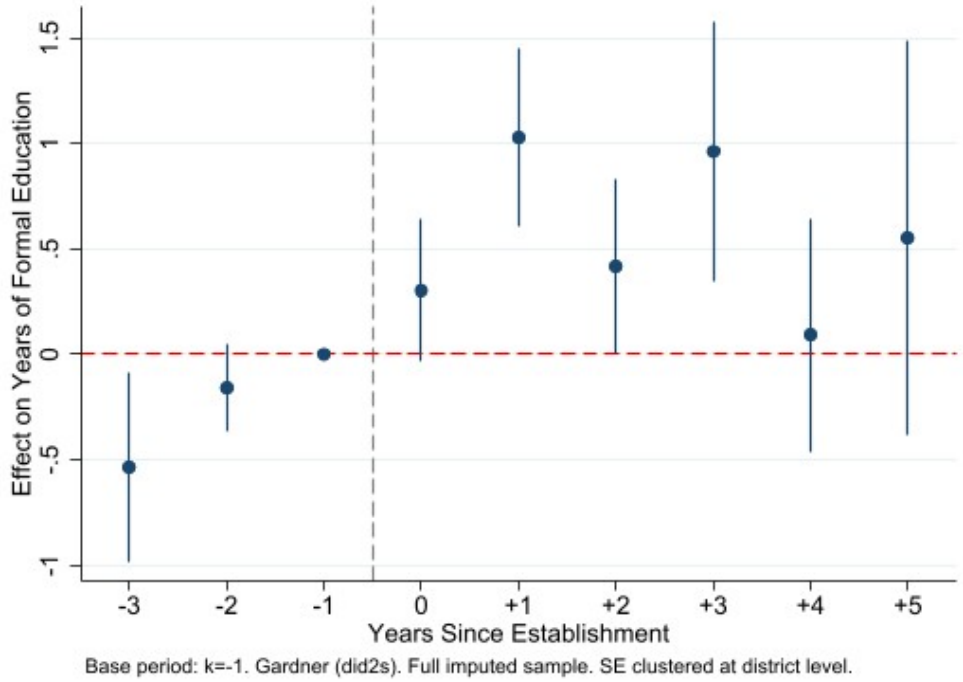
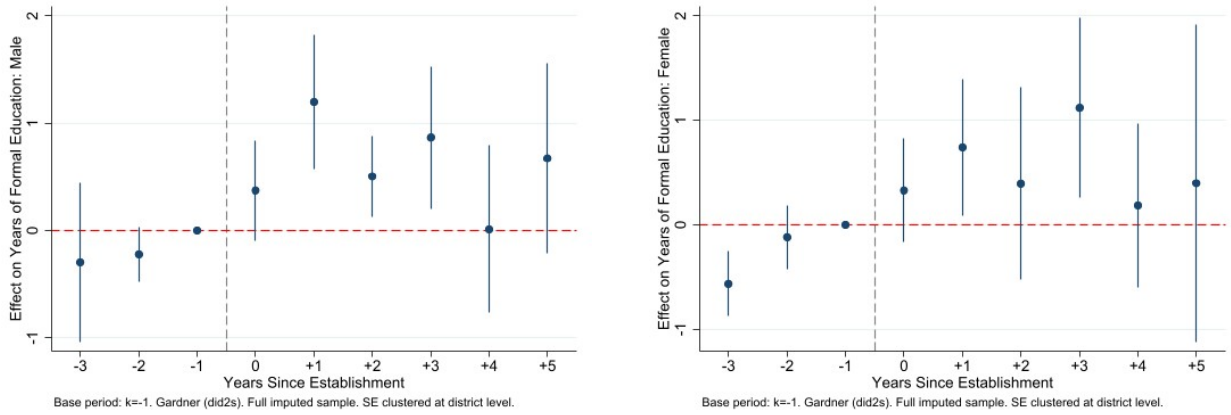


Figure 11: Years of Education at Age of Establishment 18 – All. Imputed Sample.



(a) Male (b) Female

Figure 12: Years of Education at Age of Establishment 18 – By Gender. Imputed Sample.

Age 19 at Establishment

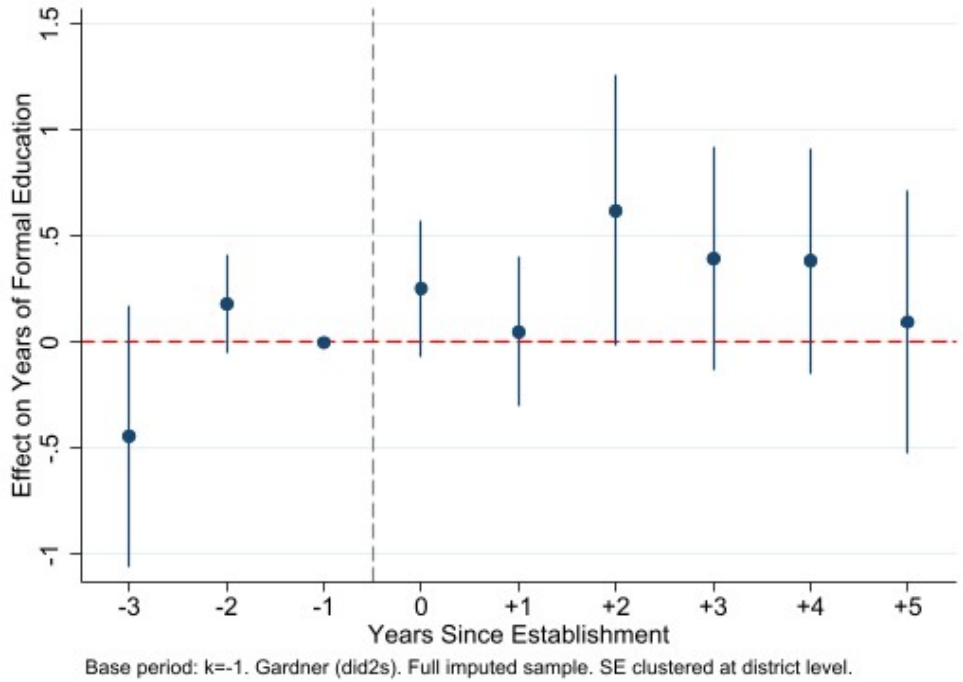
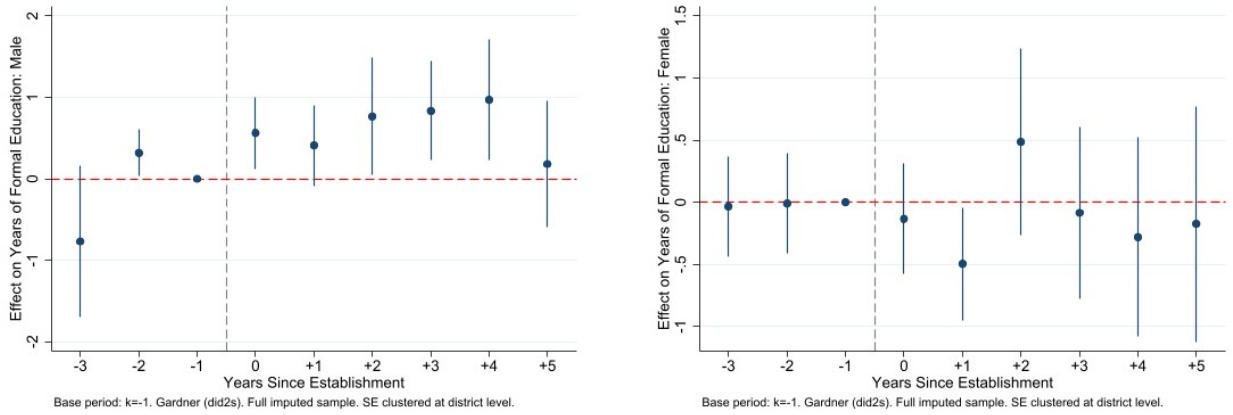


Figure 13: Years of Education at Age of Establishment 19 – All. Imputed Sample.



(a) Male (b) Female

Figure 14: Years of Education at Age of Establishment 19 – By Gender. Imputed Sample.

Graduation Rate

Next, Figure 14-17 present the event-study estimates for graduation rates using the full imputed sample. The results are broadly consistent with the baseline specification. Positive effects emerge several years after college establishment, reflecting the time required to complete a bachelor's degree, although the estimates remain relatively noisy.

The gender-specific results also follow the same pattern as the main sample. For individuals who were 18 years old at establishment, positive effects are concentrated among women, while the estimates for men remain small. For the 19-year-old cohort, positive effects are observed for both men and women. Although the magnitudes are somewhat smaller, the similarity in the timing and direction of the effects supports the robustness of the main findings.

Age 18 at Establishment

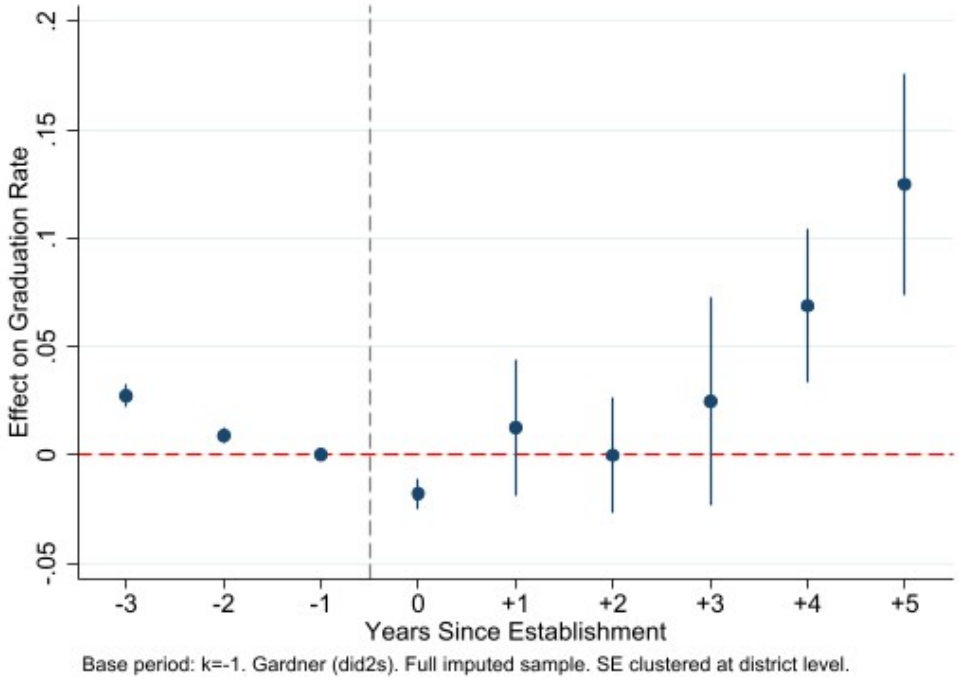
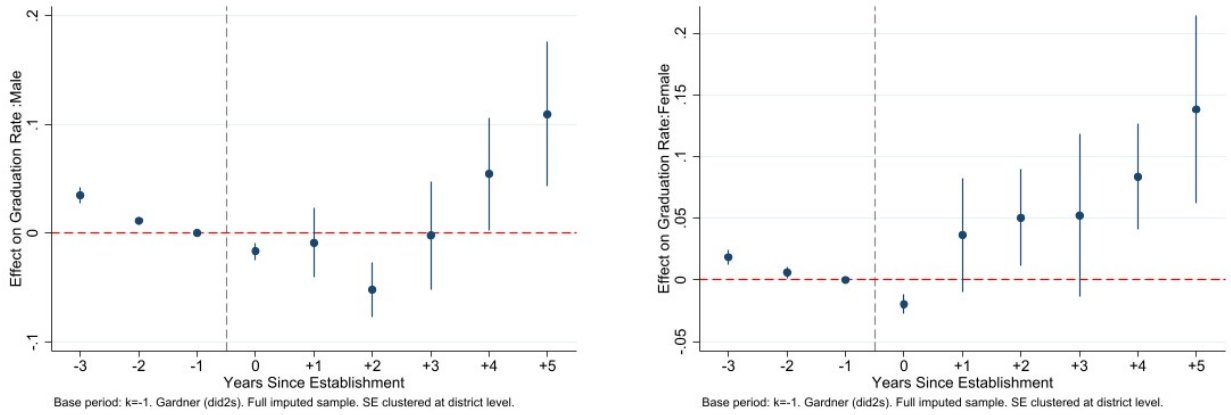


Figure 15: Graduation Rate at Age of Establishment 18 – All. Imputed Sample.



(a) Male

(b) Female

Figure 16: Graduation Rate at Age of Establishment 18 – By Gender. Imputed Sample.

Age 19 at Establishment

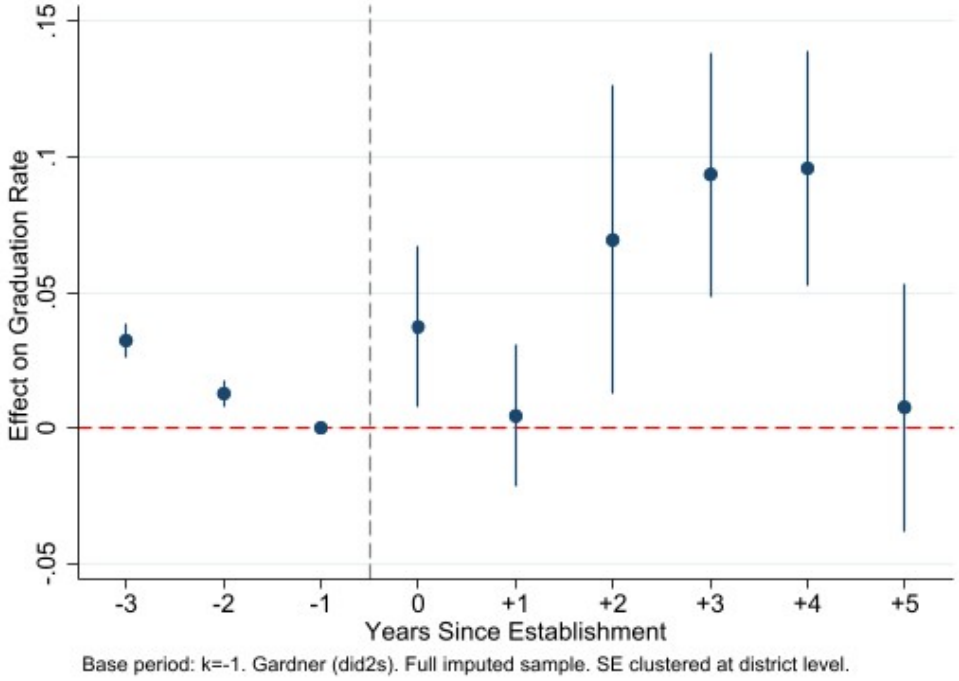
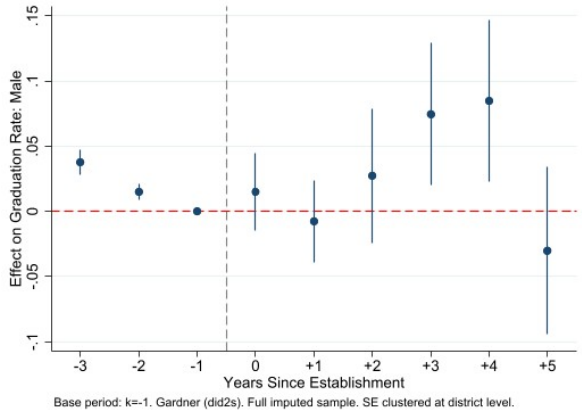
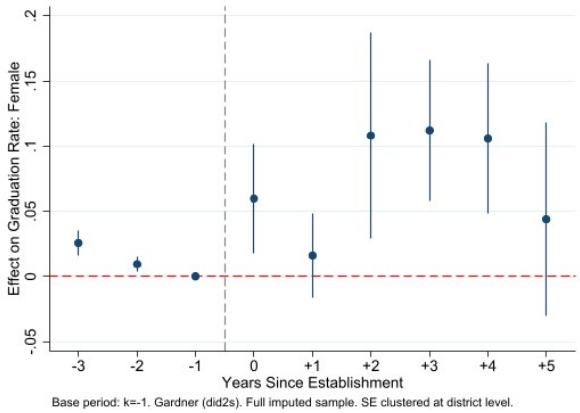


Figure 17: Graduation Rate at Age of Establishment 19 – All. Imputed Sample.



(a) Male



(b) Female

Figure 18: Graduation Rate at Age of Establishment 19 – By Gender. Imputed Sample.